Mode-Aware Anti-Jamming for Dynamic Spectrum Sharing UAV Networks:Joint Spectrum Allocation, Trajectory, and Power Optimization

Hao Zhou[§], Fuhui Zhou[§,*], and Qihui Wu[§]
[§]Nanjing University of Aeronautics and Astronautics, Nanjing, China
Email: *zhouhao1@nuaa.edu.cn, zhoufuhui@ieee.org, wuqihui2014@sina.com*

***Abstract*—Unmanned aerial vehicle (UAV) networks operating under dynamic spectrum sharing (DSS) are vulnerable to intelligent jamming, especially when adversaries adopt pattern-switching strategies that induce non-stationary interference. Such dynamics significantly degrade communication reliability and hinder fast recovery after interference transitions. To address this issue, a mode-aware anti-jamming framework for DSS-enabled UAV networks is proposed. A CNN–GRU based module is first designed to recognize jamming modes from short-term interference observations. The inferred mode information is then incorporated into a semantic-augmented Markov decision process. Based on this formulation, a hybrid reinforcement learning framework combining proximal policy optimization (PPO) and deep deterministic policy gradient (DDPG) is developed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrate that the proposed approach achieves faster adaptation to jamming mode transitions and significantly improves throughput and communication reliability compared with baseline schemes.**

***Index Terms*—Dynamic spectrum sharing, deep reinforcement learning, mode-aware anti-jamming, resource allocation**

I. Introduction

Recently, unmanned aerial vehicles (UAVs) have emerged as a promising solution for providing flexible and on-demand wireless connectivity in low-altitude networks. Benefiting from their high mobility and favorable line-of-sight (LoS) communication links, UAV-assisted systems have been widely applied in emergency communications, temporary hotspot coverage, and aerial relaying [1]–[3]. However, the limited availability of dedicated spectrum resources poses a fundamental challenge to UAV communications. To alleviate spectrum scarcity, dynamic spectrum sharing (DSS) has emerged as a promising paradigm, enabling UAV networks to opportunistically access licensed or unlicensed spectrum while coexisting with incumbent users [4]. Nevertheless, the open and dynamic nature of DSS environments makes UAV communications highly susceptible to interference, particularly in the presence of dense wireless devices and potential adversarial jamming. As a result, the wireless environment becomes highly dynamic and uncertain, posing significant challenges for efficient and adaptive resource allocation, including spectrum allocation, UAV trajectory optimization, and transmit power control.

The existing resource allocation approaches can be broadly categorized into two classes, namely, conventional optimization-based methods and intelligent learning-based methods, particularly those based on deep reinforcement learning (DRL). In [5], an alternating iterative optimization framework was developed to jointly optimize transmit power and UAV trajectory, where successive convex optimization was applied to trajectory design. In [6], the problem was extended to spectrum sharing scenarios, where joint trajectory and transmit power optimization was addressed using block coordinate descent (BCD) combined with successive convex optimization. Furthermore, more complex interference environments were considered in [7], where trajectory and power control under malicious jamming was solved via an alternating optimization (AO) framework with successive convex approximation (SCA) and semidefinite relaxation (SDR). Moreover, in [8], the problem was further generalized to mixed-integer formulations involving joint resource allocation and UAV trajectory control, which was tackled by decomposing it into subproblems and solving via alternating optimization. Despite their effectiveness, conventional optimization-based methods often lead to nonlinear and non-convex problems due to coupled variables and sensitivity to initial conditions, making them difficult to solve efficiently and prone to suboptimal solutions.

To overcome the limitations of conventional optimization-based methods, DRL-based approaches have gained increasing attention. In [9], a clustering-aided reinforcement learning approach was proposed for joint user association, UAV trajectory design, and power control. In [10], a DRL-based trajectory design scheme was developed for UAV communication systems against malicious jamming. Moreover, in [11], an online-offline MA2C-DDPG framework was proposed for dynamic spectrum sharing UAV networks, where spectrum allocation and UAV trajectory were jointly optimized. This direction was further advanced in [12], where an intelligent learning framework from external interaction to internal inference was developed for spectrum sharing and UAV trajectory optimization. In addition, in [13], a multi-agent DRL-based scheme was proposed for joint trajectory control, resource allocation, and user association in multi-fixed-wing UAV networks.

However, most existing DRL-based methods characterize interference only through observable states, such as channel conditions or received interference power, without explicitly modeling the underlying jammer behavior. For example, in [14], channel prediction was incorporated to improve resource allocation in dynamic spectrum-sharing networks,

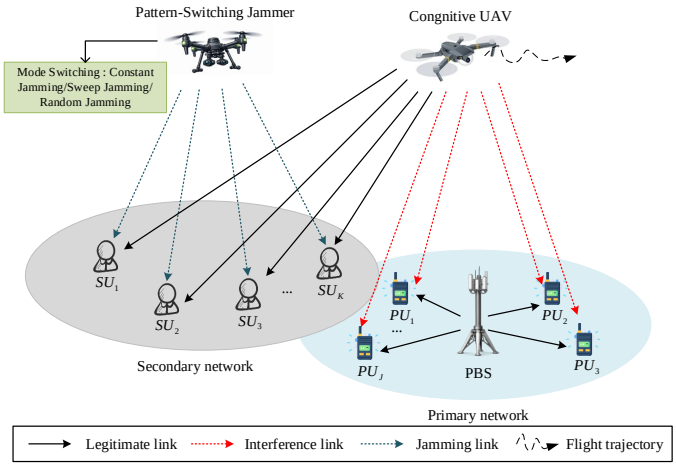

Fig. 1. Dynamic spectrum sharing UAV network under pattern-switching jamming, where the C-UAV performs jamming mode awareness and jointly optimizes spectrum allocation, trajectory, and transmit power to ensure reliable communication.

while in [15], anti-jamming strategies were designed based on power and mobility control. Nevertheless, pattern-switching jamming, where interference modes change abruptly over time, has rarely been explicitly considered, leading to slow adaptation in highly non-stationary environments.

In this paper, a mode-aware anti-jamming framework is developed for DSS-enabled UAV networks under pattern-switching interference. Unlike conventional approaches that rely on implicit interference learning, A CNN–GRU based module is first introduced to recognize jamming modes from short-term interference observations and capture the dynamic interference patterns. By integrating the inferred jamming mode into the Markov decision process, the proposed framework enables faster adaptation to non-stationary environments. Furthermore, a hybrid reinforcement learning framework combining PPO and DDPG is designed to address the coupled discrete–continuous optimization problem, where spectrum allocation, UAV trajectory, and transmit power are jointly optimized. Extensive simulation results verify that the proposed framework significantly improves communication reliability and spectrum efficiency, while achieving faster recovery after jamming mode transitions compared with baseline schemes.

The remainder of this paper is organized as follows. The system model is presented in Section II. Section III presents our proposed mode-aware anti-jamming framework. Section IV presents the simulation results. Finally, the paper concludes with Section V.

II. System Model

A. Scenario Description

We consider a UAV-enabled dynamic spectrum sharing (DSS) network under intentional pattern-switching jamming, as illustrated in Fig. 1. The network consists of a primary network and a cognitive UAV-assisted secondary network, which coexist over a set of licensed spectrum bands.

The primary network includes a primary base station and J primary users (PUs). The PUs are indexed by $j \in \mathcal{J} \triangleq \{1, 2, \ldots, J\}$. In order to serve more secondary users (SUs) and provide better services, a wideband spectrum is divided into M sub-carrier spectrum bands. The spectrum bands are indexed by $m \in \mathcal{M} \triangleq \{1, 2, \ldots, M\}$. To eliminate mutual interference among PUs, each PU is assigned an exclusive licensed spectrum band. The secondary network comprises a cognitive UAV serving K ground SUs. The SUs are indexed by $k \in \mathcal{K} \triangleq \{1, 2, \ldots, K\}$. The C-UAV operates as an aerial secondary base station, providing downlink communications to SUs by opportunistically accessing the licensed spectrum while ensuring that the interference inflicted on the primary network remains within acceptable limits.

A three-dimensional Cartesian coordinate system is adopted. The horizontal locations of the PBS, the j-th PU, the k-th SU, the C-UAV and the J-UAV at time slot n are denoted by $\mathbf{w}_b = (x_b, y_b)$, $\mathbf{w}_{p,j} = (x_{p,j}, y_{p,j})$, $\mathbf{w}_{s,k} = (x_{s,k}, y_{s,k})$, $\mathbf{q}_c = (x_c, y_c)$ and $\mathbf{q}_{jam} = (x_{jam}, y_{jam})$ respectively. These UAVs fly at a fixed altitude H_c. The total communication duration T is divided into N equal-length time slots, each with duration $\Delta t = T/N$. Due to the short duration of each slot, the C-UAV is assumed to be quasi-static within one slot. Let $n \in \mathcal{N} \triangleq \{1, 2, \ldots, N\}$ denote the set of time slots. The dynamic positions of the SUs can be formulated as

$$\begin{aligned} x_{s,k}[n+1] &= x_{s,k}[n] + v_s \cos\big(\phi_{s,k}[n]\big), \\ y_{s,k}[n+1] &= y_{s,k}[n] + v_s \sin\big(\phi_{s,k}[n]\big), \end{aligned} \tag{1}$$

where $\phi_{s,k}[n] \in [-\pi, \pi]$ denotes the direction of the k-th SU at time slot n. The SUs move at a fixed speed v_s. Similarly, the dynamic position of the C-UAV can be formulated as

$$\begin{aligned} x_c[n+1] &= x_c[n] + v_c \cos\big(\phi_c[n]\big), \\ y_c[n+1] &= y_c[n] + v_c \sin\big(\phi_c[n]\big), \end{aligned} \tag{2}$$

where v_c is the constant flight speed, and $\phi_c[n] \in [-\pi, \pi]$ denotes the direction of the C-UAV. The distance between the C-UAV and the k-th SU, and that between the PBS and the j-th PU at time slot n are respectively given as,

$$\begin{aligned} d_{c,k}[n] &= \sqrt{\big\|\mathbf{q}_c[n] - \mathbf{w}_{s,k}[n]\big\|^2 + H_c^2}, \\ d_{p,j}[n] &= \sqrt{\big\|\mathbf{w}_b[n] - \mathbf{w}_{p,j}[n]\big\|^2}, \end{aligned} \tag{3}$$

The wireless channel between the UAV and the ground users is dominated by the line-of-sight (LoS) link. Let β_{ref} represent the channel power gain at the reference distance of 1 m. Thus, the channel power gain from the C-UAV to the k-th SU can be expressed as

$$h_{C,k,n}^{\text{LoS}} = \beta_{\text{ref}} d_{c,k}[n]^{-2}, \tag{4}$$

where $d_{c,k}[n]$ denotes the distance between the C-UAV and the k-th SU at time slot n. The channel model between the PBS and the ground users is different from that between the UAV and the ground users.It is required to consider both the

distance-dependent path loss with exponent $\varphi \geq 2$ and small-scale Rayleigh fading. Thus, the channel power gains from the PBS to the j-th PU can be given as

$$h_{P,j,n}^{\mathrm{NLoS}} = \beta_{\mathrm{ref}} d_{p,j}[n]^{-\varphi} \zeta_j, \tag{5}$$

where $d_{p,j}[n]$ is the distance between the PBS and the j-th PU at time slot n. The random variable ζ_j follows an exponential distribution with unit mean, which accounts for the Rayleigh fading.

Unlike conventional fixed or random jamming models, the J-UAV is assumed to adopt a pattern-switching strategy. Let $z[n] \in \mathcal{Z} \triangleq \{\text{Constant}, \text{Sweep}, \text{Random}\}$ denote the jamming mode at time slot n, corresponding to constant jamming, sweep jamming, and random jamming, respectively. The jamming mode evolves over time according to a finite-state stochastic process. In particular, the mode transition is modeled as a first-order Markov chain, i.e.,

$$\Pr(z[n+1] = z' \mid z[n] = z) = \Pi_{z,z'}, \quad z, z' \in \mathcal{Z}. \tag{6}$$

where $\mathbf{\Pi}$ denotes the mode transition probability matrix. The transition probabilities are not known to the C-UAV, and the instantaneous jamming mode cannot be directly observed. The pattern-switching behavior of the jammer directly translates into time-varying and non-stationary interference across the spectrum.

Considering the sub-carrier allocation, the binary variables $\rho_{k,n}[m]$ and $\rho_{J,n}[m]$ are introduced to characterize the spectrum usage of the SUs and the J-UAV, respectively.Specifically, $\rho_{k,n}[m] = 1$ indicates that the m-th sub-carrier is assigned to the k-th SU at time slot n, and $\rho_{k,n}[m] = 0$ otherwise. Each PU operates on a preassigned orthogonal sub-carrier with a fixed transmit power. The J-UAV dynamically occupies sub-carriers according to its current jamming mode, which is captured by the jamming indicator $\rho_{J,n}[m]$. When $\rho_{J,n}[m] = 1$, the m-th sub-carrier is subject to intentional interference from the jammer at time slot n. Based on the sub-carrier availability and the time-varying jamming condition, the C-UAV aims to dynamically adapt the spectrum allocation for the SUs, such that the communication quality can be rapidly restored after jamming pattern changes, while opportunistically accessing the licensed spectrum and guaranteeing the quality-of-service requirements of the primary network.

The SINR of the k-th SU and the j-th PU in the m-th sub-band can be respectively expressed as

$$\gamma_{m,k,n}^{s} = \frac{\rho_{k,n}[m] P_C[n] h_{C,k,n}^{\mathrm{LoS}}}{\sigma^2 + P_B h_{P,k,n}^{\mathrm{NLoS}} + \rho_{J,n}[m] P_J h_{J,k,n}^{\mathrm{LoS}}}, \tag{7a}$$

$$\gamma_{m,j,n}^{p} = \frac{P_B h_{P,j,n}^{\mathrm{NLoS}}}{\sigma^2 + \sum_{k=1}^{K} \rho_{k,n}[m] P_C[n] h_{C,j,n}^{\mathrm{LoS}} + \rho_{J,n}[m] P_J h_{J,j,n}^{\mathrm{LoS}}}, \tag{7b}$$

where P_B and P_J denote the transmit power of the PBS and the J-UAV, respectively. $P_C[n] \in [P_C^{\min}, P_C^{\max}]$ denotes the transmit power of the C-UAV at time slot n. Moreover, σ^2 is the noise power. Then, the achievable data transmission rate of the k-th SU and the j-th PU can be respectively expressed as

$$R_{m,k,n}^{s} = B \log_2\left(1 + \gamma_{m,k,n}^{s}\right), \tag{8a}$$

$$R_{m,j,n}^{p} = B \log_2\left(1 + \gamma_{m,j,n}^{p}\right), \tag{8b}$$

where B denotes the bandwidth for each sub-band.

To capture communication reliability under pattern-switching jamming, we introduce a QoS threshold R_{th}. An outage event is declared when the achievable rate of a SU falls below R_{th}. Accordingly, the outage indicator is defined as

$$\mathbb{I}_{k,n}^{\mathrm{out}} \triangleq \mathbf{1}\left\{R_{k,n}^{s} < R_{\mathrm{th}}\right\}, \quad \forall k \in \mathcal{K},\ \forall n \in \mathcal{N}, \tag{9}$$

where $\mathbf{1}\{\cdot\}$ equals 1 if the condition holds and 0 otherwise. This definition penalizes each outage time slot; thus, longer outage durations yield larger cumulative penalties, which encourages fast recovery after jamming pattern transitions.

B. Problem Formulation

In this paper, we address a joint spectrum allocation, UAV trajectory control, and transmit power adaptation problem in a dynamic spectrum sharing UAV network under pattern-switching jamming, aiming to achieve fast recovery and reliable communication for secondary users while satisfying the primary user protection constraints. Accordingly, the optimization problem is formulated as follows

$$\textbf{P1:} \quad \max_{\boldsymbol{\rho}, \boldsymbol{\phi}, \mathbf{P}} \quad \frac{1}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} \sum_{m=1}^{M} R_{m,k,n}^{\mathrm{s}} - \frac{\lambda_{\mathrm{out}}}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbb{I}_{k,n}^{\mathrm{out}} \tag{10a}$$

$$\text{s.t. } \textbf{C1:} \quad \sum_{m=1}^{M} R_{m,j,n}^{p} \geq R_j^{\min}, \quad \forall j \in \mathcal{J},\ \forall n \in \mathcal{N} \tag{10b}$$

$$\textbf{C2:} \quad \rho_{k,n}[m] \in \{0,1\}, \quad \forall k \in \mathcal{K},\ \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{10c}$$

$$\textbf{C3:} \quad \sum_{k=1}^{K} \rho_{k,n}[m] \leq 1, \quad \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{10d}$$

$$\textbf{C4:} \quad |\phi_c[n]| \leq \pi, \quad \forall n \in \mathcal{N} \tag{10e}$$

$$\textbf{C5:} \quad P_C^{\min} \leq P_C[n] \leq P_C^{\max}, \quad \forall n \in \mathcal{N} \tag{10f}$$

To jointly account for throughput performance and communication reliability, a reliability-aware objective function is adopted. Specifically, the achievable rate of the secondary users is maximized, while a penalty term is introduced to capture QoS outages. This outage penalty discourages prolonged service interruptions caused by dynamic jamming and implicitly promotes fast recovery of communication quality after jamming pattern transitions. In problem (P1), $R_j^{\min}$ denotes the minimum transmission rate requirement of the j-th PU, and $\boldsymbol{\rho} = \{\rho_{k,n}[m]\}_{k \in \mathcal{K}, m \in \mathcal{M}, n \in \mathcal{N}}$ denotes the spectrum allocation policy for the SUs. The variable $\boldsymbol{\phi} = \{\phi[n]\}_{n \in \mathcal{N}}$ denotes the UAV heading angle control sequence, while $\mathbf{P} = \{P_C[n]\}_{n \in \mathcal{N}}$ denotes the transmit power adaptation policy of the cognitive UAV. Constraint (10b) guarantees the QoS requirements of the PUs in the dynamic spectrum sharing

environment. Constraints (10c) and (10d) ensure orthogonal sub-carrier allocation among SUs to avoid intra-secondary interference. Constraint (10e) enforces feasible UAV mobility, while constraint (10f) restricts the transmit power of the C-UAV within practical limits.

III. MODE-AWARE HYBRID PPO-DDPG FRAMEWORK FOR ANTI-JAMMING UAV COMMUNICATIONS

A. Jamming Mode Recognition

The jammer switches among finite modes $z[n] \in \mathcal{Z}$ following the Markov model in (6). The C-UAV performs spectrum sensing and collects the per-slot interference-energy vector

$$\mathbf{I}[n] = [I_1[n], \ldots, I_M[n]] \in \mathbb{R}^M, \tag{11}$$

and forms a length-L short window

$$\mathcal{I}[n] = \{\mathbf{I}[n-L+1], \ldots, \mathbf{I}[n]\} \in \mathbb{R}^{L\times M}. \tag{12}$$

A lightweight recognizer $f_\theta(\cdot)$ maps the input $\mathcal{I}[n]$ to a mode probability vector

$$\boldsymbol{\pi}[n] = f_\theta(\mathcal{I}[n]) \in \mathbb{R}^{|\mathcal{Z}|}, \quad \sum_{c=1}^{|\mathcal{Z}|} \pi_c[n] = 1, \tag{13}$$

which provides semantic information about the current jamming behavior. We adopt a CNN–GRU classifier, where a 1D CNN encodes the spectral characteristics of each $\mathbf{I}[\cdot]$, and a GRU captures short-term temporal dynamics over L time slots. The model is trained offline using the cross-entropy loss

$$\mathcal{L}_{\text{cls}}(\theta) = -\sum_{c=1}^{|\mathcal{Z}|} z_c[n] \log\big(\pi_c[n]\big), \tag{14}$$

where $z_c[n]$ denotes the one-hot encoded ground-truth jamming mode at time slot n. During online operation, the model is used solely for inference to output $\boldsymbol{\pi}[n]$.

B. Semantic-Augmented MDP Formulation

Following the MDP formulation in the reference work [11], we incorporate the mode probability vector $\boldsymbol{\pi}[n]$ as a semantic descriptor of the interference environment to facilitate fast adaptation under mode-switching jamming.

State: The system state at slot n is defined as

$$\mathbf{s}[n] = \big(\mathbf{a}[n-1],\, \mathbf{q}_c[n-1],\, \mathbf{g}[n],\, \boldsymbol{\pi}[n]\big), \tag{15}$$

where $\mathbf{a}[n-1]$ denotes the previous action at slot $n-1$, $\mathbf{q}_c[n-1]$ denotes the previous UAV position at slot $n-1$, $\mathbf{g}[n]$ denotes channel and geometry-related features for rate evaluation, and $\boldsymbol{\pi}[n]$ encodes the high-level semantic belief of the current jamming behavior.

Action: The action consists of a discrete spectrum decision and a continuous control vector,

$$\mathbf{a}[n] = \big(\mathbf{a}_d[n],\, \mathbf{a}_c[n]\big), \tag{16}$$

The discrete action $\mathbf{a}_d[n]$ represents a joint subcarrier allocation for all K SUs at slot n, i.e.,

$$\mathbf{a}_d[n] = \big(a_1[n],\, a_2[n],\, \ldots,\, a_K[n]\big), \tag{17}$$

where $a_k[n] \in \{1, 2, \ldots, M\}$ denotes the selected subcarrier index for the k-th SU. Equivalently, $\mathbf{a}_d[n]$ can be mapped to the binary allocation variables $\{\rho_{k,n}[m]\}$ satisfying $\sum_{m=1}^{M} \rho_{k,n}[m] = 1$ for each k. The continuous action $\mathbf{a}_c[n]$ is defined as

$$\mathbf{a}_c[n] = \big(\phi_c[n],\, P_C[n]\big), \tag{18}$$

where $\phi_c[n] \in [-\pi, \pi]$ and $P_C[n] \in [P_C^{\min},\, P_C^{\max}]$.

Reward: Consistent with (10), we adopt a reliability-aware reward that jointly captures the secondary throughput, outage reliability, and PU protection, defined as

$$r[n] = \alpha \sum_{k=1}^{K} R_k^s[n] + \beta \sum_{j=1}^{J} \delta_j[n] - \gamma \sum_{k=1}^{K} \mathbb{I}(R_k^s[n] < R_{\text{th}}), \tag{19}$$

where $\alpha, \beta, \gamma > 0$ are constant coefficients, $\delta_j[n]$ is a penalty term when the interference to the PUs exceeds the tolerated threshold and formulated as

$$\delta_j[n] = \begin{cases} 0, & R_j^p[n] \geq R_j^{\min}, \\ R_j^p[n] - R_j^{\min}, & 0 \leq R_j^p[n] < R_j^{\min}, \end{cases} \tag{20}$$

where $R_j^p[n]$ is the achievable rate of the j-th PU and $R_j^{\min}$ is its minimum rate requirement.

C. Hybrid PPO-DDPG Framework

Problem (10) involves discrete spectrum allocation, continuous joint control of UAV mobility and transmit power. Thus, we employ a hybrid solver, as shown in Fig. 2, where PPO is adopted for the discrete spectrum decision $\mathbf{a}_d[n]$ and DDPG is used for the continuous control vector $\mathbf{a}_c[n]$, both conditioned on the semantic-augmented state $\mathbf{s}[n]$.

PPO for Spectrum Allocation: Let $\pi_\theta(\mathbf{a}_d|\mathbf{s})$ denote the stochastic policy for spectrum allocation parameterized by θ, where $\mathbf{a}_d[n]$ is the discrete joint subcarrier allocation decision at time slot n. Different from the multi-agent A2C design in [11], we adopt a centralized PPO-based spectrum allocator. To avoid the exponential growth of the joint discrete action space, the policy is implemented in a factorized multi-discrete form as

$$\pi_\theta(\mathbf{a}_d|\mathbf{s}) = \prod_{k=1}^{K} \pi_\theta^{(k)}\big(a_k|\mathbf{s}\big), \tag{21}$$

where each $\pi_\theta^{(k)}(\cdot|\mathbf{s})$ is a categorical distribution over the M candidate subcarriers. To satisfy the orthogonal subcarrier constraint, action masking is adopted during action selection to prevent infeasible allocations and ensure that each subcarrier is assigned to at most one SU per slot. The objective of PPO is to maximize the expected discounted return while preventing excessively large policy updates. Specifically, PPO optimizes the clipped surrogate objective

$$\begin{aligned} \mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_n\Big[\min\Big\{ & r_n(\theta)\hat{A}_n, \\ & \text{clip}(r_n(\theta),\, 1-\epsilon,\, 1+\epsilon)\, \hat{A}_n \Big\}\Big], \end{aligned} \tag{22}$$

where $r_n(\theta) = \frac{\pi_\theta(\mathbf{a}_d[n]|\mathbf{s}[n])}{\pi_{\theta_{\text{old}}}(\mathbf{a}_d[n]|\mathbf{s}[n])}$ is the probability ratio between the new and old policies, ϵ is the clipping parameter, and

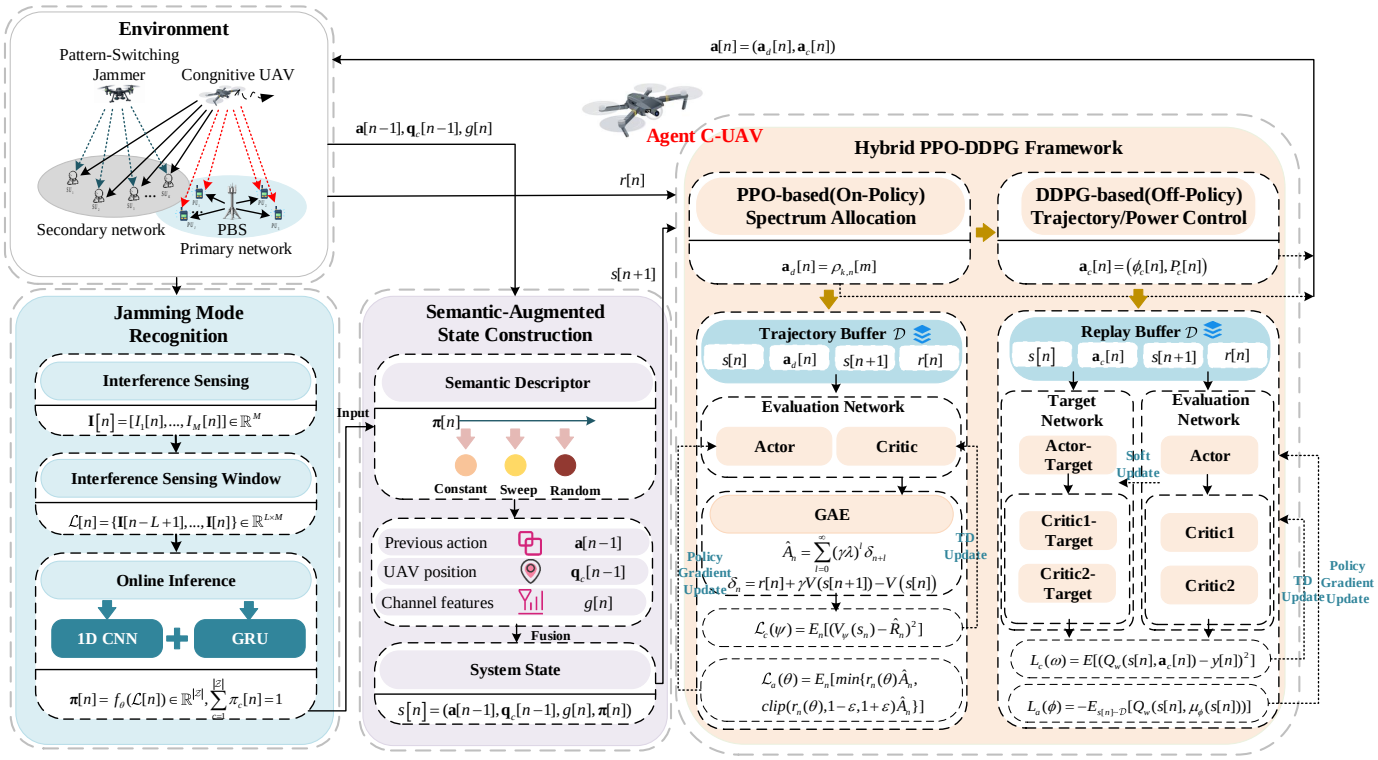

Fig. 2. The mode-aware hybrid PPO-DDPG framework for joint spectrum allocation, UAV trajectory, and power control.

$\hat{A}_n$ denotes the advantage estimate. To reduce variance and improve sample efficiency, the advantage is computed using Generalized Advantage Estimation (GAE),

$$\hat{A}_n = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{n+l}, \quad \delta_n = r[n] + \gamma V(\mathbf{s}[n+1]) - V(\mathbf{s}[n]), \tag{23}$$

where $V(\cdot)$ is the value function, γ is the discount factor, and λ controls the bias–variance tradeoff.

DDPG for Trajectory/Power Control: Given the semantic-augmented state $\mathbf{s}[n]$, we employ DDPG to learn the continuous control policy in an off-policy manner. Specifically, the transition tuple $(\mathbf{s}[n], \mathbf{a}_c[n], r[n], \mathbf{s}[n+1])$ is stored in a replay buffer, and mini-batches are uniformly sampled to update the critic network $Q_\omega(\mathbf{s}, \mathbf{a}_c)$. For each sampled transition, the bootstrap target is constructed by the target networks as

$$y[n] = r[n] + \gamma\, Q_{\bar{\omega}}\Big(\mathbf{s}[n+1],\, \mu_{\bar{\varphi}}(\mathbf{s}[n+1])\Big), \tag{24}$$

and the critic is trained by minimizing the temporal-difference loss

$$L_c(\omega) = \mathbb{E}\Big[\big(Q_\omega(\mathbf{s}[n], \mathbf{a}_c[n]) - y[n]\big)^2\Big]. \tag{25}$$

The actor is then updated to maximize the critic-evaluated Q-value, while the corresponding target networks are updated via Polyak averaging. At each time slot, PPO outputs the discrete spectrum allocation $\mathbf{a}_d[n]$ (on-policy), and DDPG outputs the continuous control action $\mathbf{a}_c[n]$ (off-policy) based on $\mathbf{s}[n]$.

IV. SIMULATION RESULTS

In this section, simulation results are presented to evaluate the performance of the proposed joint spectrum allocation, trajectory optimization, and power control scheme. The simulation settings are based on the works in [11]. Specifically, SUs are PUs are randomly deployed in two seperate $500\,\mathrm{m} \times 500\,\mathrm{m}$ cells with a C-UAV flying at a fixed altitude of $100\,\mathrm{m}$. The number of SUs is 4. The transmit power of C-UAV ranges from 200mW to 300mW. The sensing window length for mode recognition is $L = 5$. The transition matrix is set as

$$\mathbf{\Pi} = \begin{bmatrix} 0.85 & 0.10 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.10 & 0.10 & 0.80 \end{bmatrix}. \tag{26}$$

Two baseline schemes are introduced for performance comparison. **MA2C-DDPG:** MA2C and DDPG are used for spectrum allocation and trajectory optimization, with the transmit power of the C-UAV fixed at 250 mW. **Hybrid PPO-DDPG:** Hybrid PPO-DDPG framework without the jamming mode recognition module.

As shown in Fig. 3, the convergence comparison of the proposed scheme with the baseline schemes is presented over the training episodes. It is seen that all three schemes exhibit an overall increasing trend and gradually converge, while noticeable reward fluctuations appear in the first 150–200 episodes due to random exploration and jammer mode switching. After around 400 episodes, the training curves become relatively stable, indicating that effective anti-jamming policies have been learned. Moreover, the proposed scheme converges

faster and reaches the highest reward among all compared methods, since the introduced mode-aware state representation reduces the ambiguity caused by non-stationary interference and makes policy learning more efficient. In contrast, Hybrid PPO-DDPG performs worse than MA2C-DDPG, showing that increasing algorithmic complexity alone is insufficient when the interference dynamics are not effectively captured.

Fig. 4 shows the average instantaneous SU sum rate under a representative jammer mode-switching sequence, where the rate drop and recovery time around each transition point are further used to quantify the transient adaptation capability. It is observed that the proposed scheme maintains the highest sum-rate level over most time slots and suffers less performance degradation after jammer mode transitions, while recovering more quickly than the benchmark schemes. This is because the proposed framework can infer the underlying jammer pattern from short-term observations, allowing the agent to adjust its spectrum-access behavior promptly when the interference mode changes. Therefore, the proposed scheme exhibits stronger robustness and faster adaptation in pattern-switching jamming environments.

Fig. 5 shows the overall performance comparison of the proposed scheme with the baseline schemes in terms of average SU sum rate and outage probability. It is seen that the proposed scheme achieves the highest average SU sum rate and the lowest outage probability among all compared methods. This indicates that the proposed mode-aware framework can make more effective long-term decisions under dynamic interference, thereby improving spectral efficiency while maintaining more reliable communication links. Therefore, compared with Hybrid PPO-DDPG and MA2C-DDPG, the proposed scheme achieves a better throughput–reliability tradeoff in DSS-enabled UAV networks.

Fig.6 shows the robustness comparison of different schemes under low, medium, and high jammer mode-switching rates. It is seen that the performance of all schemes degrades as the switching rate increases, while the proposed scheme consistently maintains the highest average SU sum rate and the lowest outage probability. Moreover, the performance degradation of the proposed scheme is smaller than that of the benchmark schemes, indicating that its advantage becomes more evident in highly non-stationary interference environments. This is because the proposed framework can better exploit jammer-mode information to track the changing interference patterns. Therefore, the proposed mode-aware scheme exhibits stronger robustness against frequent jammer mode transitions.

V. Conclusion

This paper investigated mode-aware anti-jamming communication in dynamic spectrum sharing UAV networks under pattern-switching interference. A mode-aware anti-jamming framework was developed in which the UAV first recognizes the current jamming pattern using a CNN–GRU based classifier and then incorporates the inferred mode information into the decision-making process. By formulating a semantic-augmented MDP, a hybrid PPO–DDPG reinforcement learning algorithm was proposed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrated that the proposed approach can effectively adapt to dynamic interference environments, achieve faster recovery after jamming mode transitions, and significantly improve both average throughput and communication reliability compared with baseline schemes. The proposed framework provides a promising solution for robust UAV communications in adversarial spectrum-sharing environments.

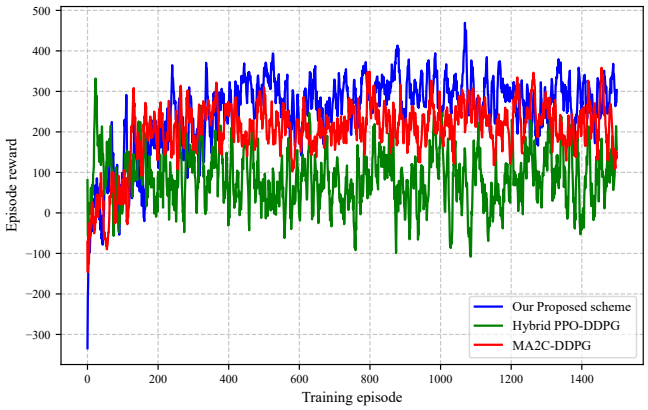

Fig. 3. Convergence comparisons of our proposed scheme with baseline schemes.

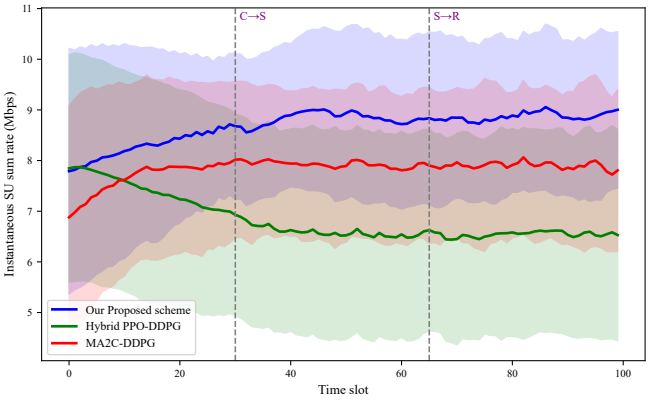

Fig. 4. Recovery comparisons of our proposed scheme with baseline schemes under jammer mode transitions.

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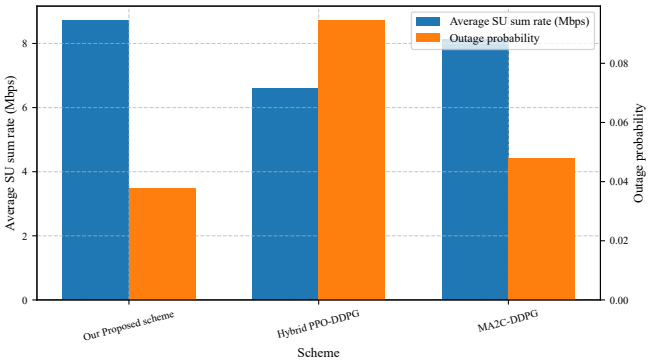

Fig. 5. Overall throughput and reliability comparisons of our proposed scheme with baseline schemes.

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